Phloem fibers: Sclerenchymatous cells associated with primary and secondary phloem are commonly called phloem fibers. These cells are elongated, lignified and provide mechanical strength to the plant body.

Table 18.2 Differences between Meristematic tissue and Permanent tissue.

Meristematic tissue	Permanent tissue
Cell wall is thin and elastic.	Cell wall is thick.
Component cells are small, spherical or polygonal and undifferentiated.	Component cells are large, differentiated with different shapes.
Cytoplasm is dense, and vacuoles are nearly absent.	Usaually large central vacuole is present in living permanent cells.
Intercellular spaces absent.	Intercellular spaces present.
Nucleus is large and prominent.	Nucleus is less conspicuous.
Cells grow and divide regularly.	Cells do not normally divide.
Provides mechanical support and elasticity to the plant body.	Provides only mechanical support .

18.2 Animal Tissues

An assemblage of one or more types of specialized cells held together with extracellular material constitute the tissue. The study of tissues is known as **Histology**.

Simple tissue: A group of cells that are similar in origin, form, structure and work together to perform a specific function.

Compound tissue: A group of cells different in their structure and function but co-ordinate to perform a specific function

Animal tissues can be grouped into four basic types on the basis of their structure and functions.

a. Epithelial tissue. b. Connective tissue
c. Muscular tissue d. Nervous tissue

18.2.1 Epithelial Tissues

It is the simplest tissue composed of one or more layers of cells covering the external surface of the body and internal organs. The cells are arranged very close to each other with less extracellular material. Epithelial cells lie on a non-cellular basement membrane. The epithelial tissue generally lacks blood vessels. The epithelium is separated by the underlying connective tissue which provides it with nutrients. There are two types of epithelial tissues. **Simple epithelium** is composed of single layer of cells resting on a basement membrane. **Compound epithelium** is composed of several layers of cells. Only the cells of the deepest layer rest on the basement membrane.

Functions of epithelial tissues

i. The skin which forms the outer covering of the body protects the underlying cells from drying, injury and microbial infections.
ii. They help in absorption of water and nutrients.
iii. They are involved in elimination of waste products.
iv. Some epithelial tissues perform secretory function (Secretion of sweat, saliva, mucus and enzymes).

A. Simple Epithelium

It is formed of **single layer** of cells. It forms a lining for the body cavities and ducts. Simple epithelium is further divided into following types.

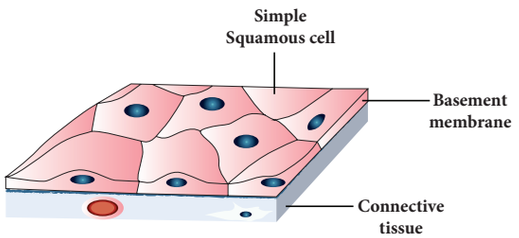


Figure 18.7 Squamous Epithelium

IX_SCI_EM_Unit-18_BIO.indd 214 06-02-2020 13:10:26